

Course: **Applied Machine Learning (reading course)** ¹

Course Code: **DA342**

Course description:

DA342 will explore some Machine Learning (ML) techniques to solve different practical problems that occur in one of the area of *Audio, Text, Image, and Video processing* task. We will start with the definition of the particular problem and some initial results. Then try to touch some state-of-the-art results from the papers published in the last *ten* years in the top venues like ICML, NeurIPS, CVPR, ICCV, ECCV and related journals.

Prerequisite (s):

Student should have some knowledge in

- Mathematics: Linear Algebra, Multivariate Calculus, Basic Optimization, Basic Probability and Statistics
- Computer programming: Any one from C/C++/**Python(recommended for the class project)**/MATLAB/Octave
- Basis ML concepts (classification, regression, etc.)

Credit Hours: 4, approximately 60 credit hours

Text(s):

Depend on the selected topics and here are some ML and related text books:

- Kevin Patrick Murphy, *Probabilistic Machine Learning: An Introduction*, MIT Press, 2021. [online]
- Christopher M. Bishop, *Pattern Recognition and Machine Learning*, Springer, 2006. [online]
- Trevor Hastie, Robert Tibshirani and Martin Wainwright, *Statistical Learning with Sparsity: The Lasso and Generalizations*, CRC Press, 2015. [online]
- R. O. Duda, P. E. Hart and D. G. Stork, *Pattern Classification and Scene Analysis*, 2nd ed., Wiley, New York, 2000
- Mehryar Mohri, Afshin Rostamizadeh, and Ameet Talwalkar, *Foundations of Machine Learning*, MIT Press, Second Edition, 2018. [online]
- Shai Shalev-Shwartz and Shai Ben-David, *Understanding Machine Learning: From Theory to Algorithms*, Cambridge University Press, 2014. [online]
- Dan Jurafsky and James H. Martin. *Speech and Language Processing*. draft, 3rd edition, 2023. [online]
- Ian Goodfellow, Yoshua Bengio, and Aaron Courville. *Deep Learning*. MIT Press, 1st edition, 2016. [online]

Course Objective:

Students will gain knowledge of:

- critical reading and analysis skills for different practical applications of machine learning algorithms.
- identifying main concepts, key arguments, evaluating evidence, and understanding the author's perspective and biases.
- recognizing the limitations of machine learning algorithms in addressing real-world problems.

Course outcomes:

After successful completion of this course, students will be able to:

- decompose a data related problem (in the area of *Audio, Text, Image, and Video processing*) into subproblems in ML and identify potential solutions.
- apply state-of-the-art ML tools and techniques to solve data related problems that occur in one of the area of *Audio, Text, Image, and Video processing* task.
- start critically read, assess, and synthesize scientific literature in applied ML and related fields.
- discuss the potential and limitation of ML models in a particular area.

Grade Distribution:

Paper reading and presentation	40%
Project and report writing	50%
Paper/Project review	10%

For paper reading and presentation, the student will be evaluated each week. The grade will be based on how much the student has understood the allocated paper/material, their presentation skills, new thinking direction and Q/A.

The **mid-term** evaluation will be based on a presentation (whatever learned so far) and a project proposal submission. Students have to define one project proposal before the midterm. The student should finalise the project proposal with the help of the instructor.

The **end-term** evaluation will be on the project. For that, students have to submit a formal project report and give a public seminar.

Apart from the mid-term and end-term students will be evaluated each week continuously. In the **Paper/Project review** section, the instructor will exchange the project report with the student and will ask them to review their friend's project report (the instructor will also do that). Based on that, students will be evaluate.

Course outline (tentative) and syllabus:

There is no predefined(fixed) syllabus for this course! The instructor will give some topics based on the current hot research topic and industry problems with a brief introduction. Students have to choose one topic based on their interests. The instructor will share some materials related to the selected topics. Each student has to present those at least once a week and have to do a project on that or a related topic.

In the past, students have selected the following topics:

- **Fake news/misinformation detection.**
- **Tackling climate change with machine learning.**
- **Player/team performance prediction in IPL game.**

- Sparse coding/learning and its application in images and video data.
- Mental Health Prediction Using Machine Learning: Assessing the Impact of Imbalance Handling Techniques.
- Similarity Searching and QSAR Models interpretation in Drug Discovery

The weekly coverage might change as it depends on the progress of the class. However, you must keep up with the reading assignments. Each week assumes 4 hour lectures. Reading assignments are from reference textbook will be mentioned in the suggested reading material in the *course url*.

Week	Content
Week 1	• Introduction to the course • Course logistics • A brief discussion of some practical problems (research and industry)
Week 2	• A brief discussion of more practical problems (research and industry) and asked the student to explore some focused workshops happening in the top venues like ICML, NeurIPS, CVPR, ICCV, ECCV, ICLR etc.. • Assignment of 2-3 topics to each student.
Week 3	• Particular topic selection for each student – Student has to give a brief presentation on the topics allocated in the last week
Week 4	• Student presentation
Week 5	• Student presentation
Week 6	• Student presentation • Finalise the class project (through a public presentation)
Week 7	• Student presentation and project update
Week 8	• Student presentation and project update
Week 9	• Student presentation and project update
Week 10	• Student presentation and project update
Week 11	• Student presentation and project update
Week 12	• Student presentation and project update
Week 13	• Student presentation and project update – Discussion on project report
Week 14	• Student presentation and project update – Review of the project report
Week 15	• Final presentations (public)

¹This is an initial draft and will update based on the suggestions/feedback.